

MEP summary: Modeling chromosome spatial organization with Self-Consistent Field Theory of Polymers

In this project, we implemented a two-part computational pipeline that connects the Hi-C maps to chromosomal spatial distributions. The first part uses Monte Carlo method to infer the chromatin “binding-site” sequence (figure 1) which represents the effective interaction pattern along the locus that gives rise to the Hi-C map. The second part uses self-consistent field theory (SCFT) of polymers to predict the spatial distribution of a chromatin polymer, with the sequence inferred in the first part.

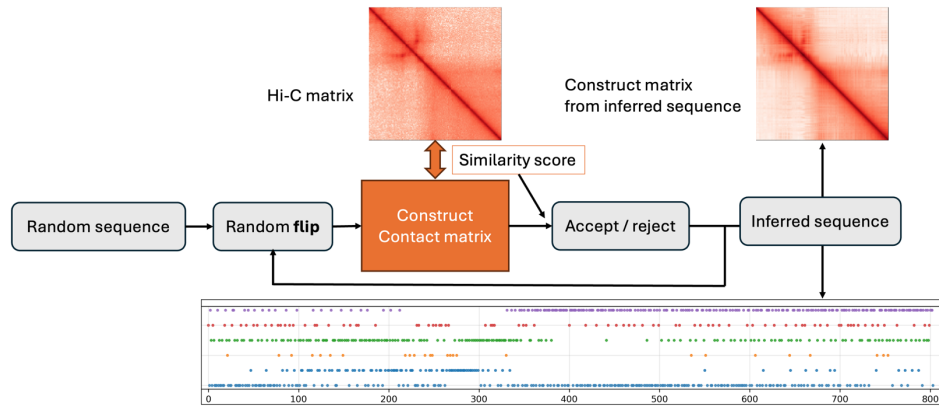


Figure 1: Workflow of the Monte Carlo process that infers the binding-site sequence. The predict contact matrix constructed from the inferred sequence the resembles closely the Hi-C map.

The SCFT translates pairwise interactions between polymer segments into a chemical potential field and its conjugated polymer probability-density distribution (Figure 2a). In our model, the chromatin (2M bp) is represented as a Gaussian chain, and the field is calculated from the Flory-Huggins mixing of the polymer segment types obtained from the part I. From the SCFT solution, we computed the polymer density and sampled a representative single-chain conformation by following the positions of maximum polymer contour density (Figure 2c, d). The contact map (contour density overlap map) of this conformation shows a high similarity to the experimental median-distance map (Figure 2b).

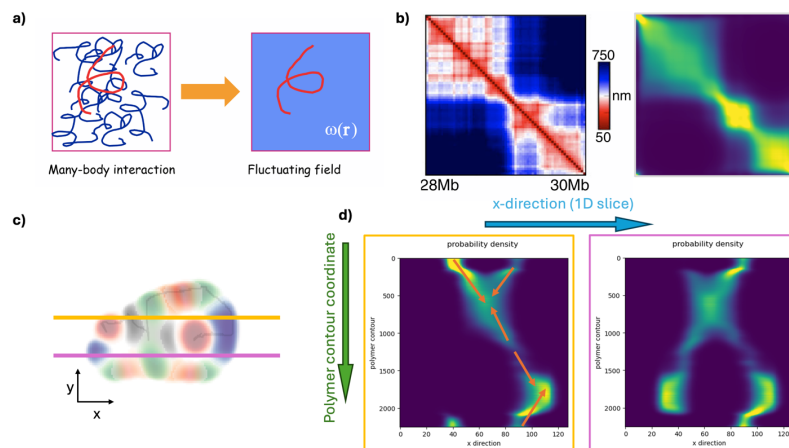


Figure 2: **a)** Schematic representation of SCFT framework. **b)** Comparison of median-distance map (left) and the contact map calculate from the SCFT result (right). **c)** Polymer density and most-probable single chain conformation (dark line). **d)** One-dimensional slices of the contour probability density. The positions of the two slices are indicated in (c).